Supplementary Information

S1. Biological parameters

Table 1: Biological parameters used in the multistage CTMC carcinogenesis model.

Parameter	Description	Value	Justification
K	Crypt carrying capacity	1500	Biological scaling assumption
β_0	Type-specific birth rate	0.15 per week	Model calibration
δ_0	Apoptosis/extrusion rates	0.14 per week	Literature-based calibration
μ_{APC}	APC mutation transition rate	0.000001	Reported mutation estimates
μ_{KRAS}	KRAS mutation transition rate	0.000015	Reported mutation estimates
μ_{TP53}	TP53 mutation transition rate	0.0000011	Reported mutation estimates

S2. Numerical settings

Table 2: Numerical settings used for the Moment–SOS optimization procedure.

Parameter	Description	Setting
d	Relaxation order	2–5
ϵ	Convergence tolerance	10^{-3}
Solver	SDP optimization solver	MOSEK/SDPT3
Platform	Computational environment	MATLAB + YALMIP

S3. SSA configuration

Table 3: Gillespie SSA benchmark configuration.

Parameter	Description	Setting
N	Number of SSA realizations	100000
T	Simulation horizon	15 years
Seed	Random seed control	Fixed seed
Output	Estimated quantities	Transformation probability and first-passage statistics

S4. Benchmark settings

Table 4: Computational environment and settings used for the Moment–SOS and SSA comparison.

Item	Description	Specification
Hardware	Processor	Intel Core i7
	RAM	16 GB
Operating system	Platform	Windows 11
Software	Programming environment	MATLAB R2024b
Optimization framework	Modeling toolbox	YALMIP
SDP solver	Semidefinite optimization solver	MOSEK
Relaxation order	Moment–SOS hierarchy levels evaluated	$d = 1, 2, 3, 4$
Final relaxation order	Order used for SSA comparison	$d = 4$
SSA realizations	Number of Gillespie trajectories	$N = 100000$

S5. Reproducibility Statement

All biological parameters, model assumptions, numerical settings, and simulation configurations used in this study are summarized in Supplementary Tables S1–S4. These details are provided to improve the reproducibility of the proposed Moment–SOS computational framework.